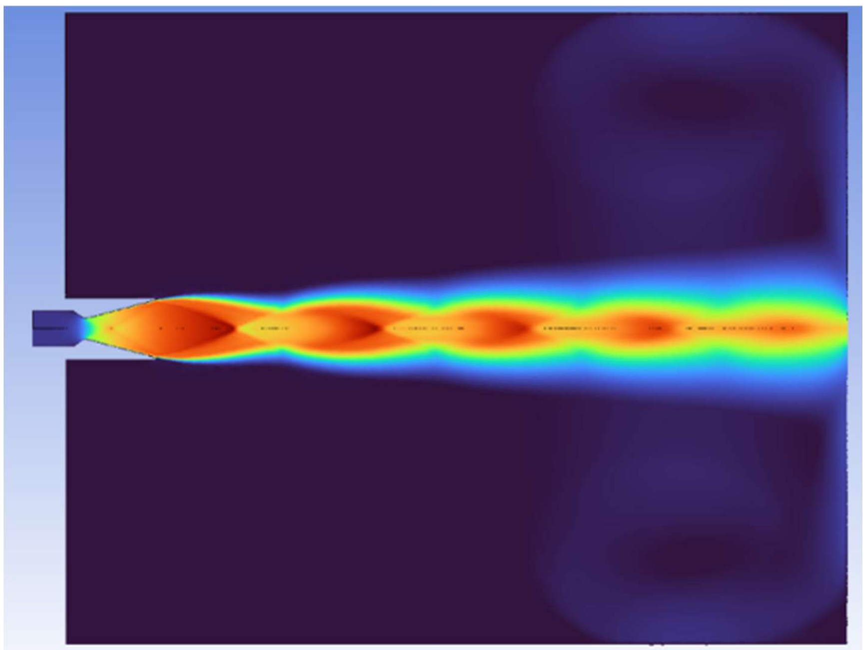
MULTIPHYSICS ANALYSIS OF A REPRESENTATIVE SMALL TURBOPUMP-FED LIQUID C-D ROCKET NOZZLE

Technical Report on High-Speed Gas Dynamics, Convective Heat Transfer, and FEA Stress Analysis



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1.ABSTRACT

The multiphysics analysis of a representative small turbopump-fed liquid convergent-divergent rocket nozzle was conducted to evaluate high-speed flow dynamics, thermal loading, and structural integrity. Initial design parameters and thermodynamic baselines were established using RPA at a 30 bar chamber pressure. A transient, 2D-axisymmetric pressure-based CFD simulation using k - ω SST turbulence model was performed to mapped the supersonic gas dynamics, pressure distributions, and shock diamond structures. Subsequently, a coupled thermal-structural FEA was executed on a 3D geometry for an uncooled Inconel 718 shell with a 4 mm wall thickness. The following literature sources on "Numerical Simulation of Supersonic Flow Through a Convergent-Divergent Nozzle", "A Simple Equation for Rapid Estimation of Rocket Nozzle Convective Heat Transfer Coefficients", and "Thermal Analysis and Design Optimization of Inconel 718 Convergent-Divergent Nozzle for High-Performance Propulsion System" were referenced. The workflow validated structural margins against operating thermal-stress state.

2. NOMENCLATURE & ACRONYMS

S. No.	Symbol / Acronym	Description
1.	CFD	Computational Fluid Dynamics
2.	FEA	Finite Element Analysis
3.	RPA	Rocket Propulsion Analysis
4.	C-D	Convergent-Divergent
5.	SST	Shear Stress Transport
6.	FoS	Factor of Safety
7.	Pc	Chamber Pressure
8.	M	Mach Number
9.	Tmax	Maximum Inner Wall Equilibrium Temperature
10.	q''	Heat Flux
11.	σ_{yield}	Material Yield Strength
12.	σ_v	Equivalent (von Mises) Stress
13.	Vj	Exhaust Jet Velocity
14.	hg	Convective Heat Transfer Coefficient
15.	CAD	Computer-Aided Design
16.	g	Acceleration due to gravity (9.81)

3. BASELINE DESIGN PARAMETERS

1) Propellants: LOX (Liquid Oxygen) + RP1 (Rocket Propellant-1)
2) Chamber Total Pressure: 30 bar
3) Nozzle: Conical Nozzle (Divergent half angle = 15°)
4) Material: Inconel 718 (for Structural & Thermal Analysis)
5) Mixture Ratio = $\frac{\dot{m}_{\text{LOX}}}{\dot{m}_{\text{RP-1}}}$ = 2.784 (Fuel Rich)

NOTE: Mixture Ratio = 2.784 (fuel rich) was used as it gives best balance of high propulsion efficiency and lower thermal loading due to lower average atomic weight of exhaust.

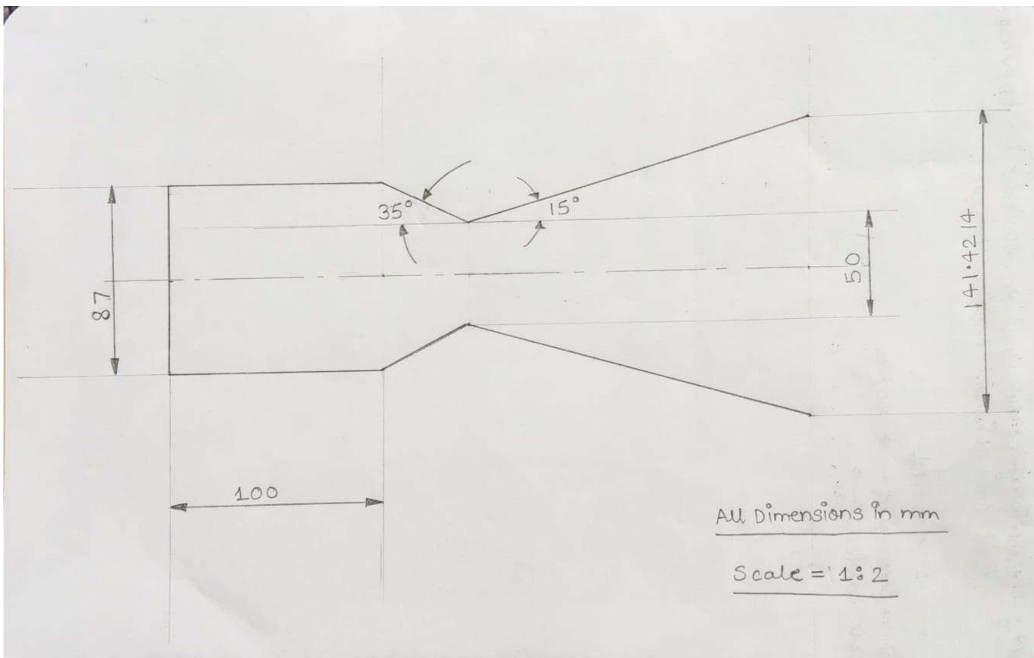
The Reaction Equation:

$$CH_{1.953} + x\,O_2 \longrightarrow a\,CO_2 + b\,H_2O + c\,CO + d\,OH + e\,H_2 + f\,O$$

Exploded Propellant Formula: $C_{0.451}H_{0.880}O_{1.097}$

Molecular Weight of RP-1 = 13.98 g/mol
Molecular Weight of LOX = 31.998 g/mol

6) LOX: 90 K, 35 bar; RP-1: 289 K, 35 bar — When kept in Propellant Tanks/Chambers (Data taken from NASA – "Guidelines for Parametric Power Cycle Study Document (Table 3)")



Basic Design of C-D Rocket Nozzle

7) Diameter of Throat = 50 mm
8) Ac / At = 3.0
   Diameter of Cylindrical Combustion Chamber = 87 mm
9) Ae / At = 8.0
   Diameter of Nozzle Exit = 141.4214 mm

4. THERMODYNAMIC PERFORMANCE ANALYSIS USING RPA

All baseline parameter data were given to RPA.

RPA Results:

Table 1. Thermodynamic properties

Parameter	Injector	Nozzle inlet	Nozzle throat	Nozzle exit	Unit
Pressure	3.0000	2.8670	1.7004	0.0480	MPa
Temperature	3567.6589	3557.2044	3401.8086	1873.6696	K
Enthalpy	-767.1752	-795.9817	-1425.8091	-4606.6377	kJ/kg
Entropy	11.3806	11.3883	11.3883	11.3883	kJ/(kg·K)
Internal energy	-2012.9805	-2037.7161	-2596.4212	-5244.3967	kJ/kg
Specific heat (p=const)	7.7011	7.7247	7.6901	1.8289	kJ/(kg·K)
Specific heat (V=const)	6.4800	6.5021	6.5230	1.4885	kJ/(kg·K)
Gamma	1.1884	1.1880	1.1789	1.2287	
Isentropic exponent	1.1297	1.1294	1.1253	1.2287	
Gas constant	0.3492	0.3491	0.3441	0.3404	kJ/(kg·K)
Molecular weight (M)	23.8105	23.8185	24.1619	24.4271	
Molecular weight (MW)	0.02381	0.02382	0.02416	0.02443	
Density	2.4081	2.3088	1.4526	0.0752	kg/m³
Sonic velocity	1186.3182	1184.2187	1147.7235	885.2095	m/s
Velocity	0.0000	240.0272	1147.7235	2771.0874	m/s
Mach number	0.0000	0.2027	1.0000	3.1304	
Area ratio	3.0000	3.0000	1.0000	8.0000	
Mass flux	554.1852	554.1852	1667.1492	208.3936	kg/(m²·s)
Mass flux (relative)	1.847e-04	1.933e-04			kg/(N·s)

Table 3. Theoretical (ideal) performance

Parameter	Sea level	Optimum expansion	Vacuum	Unit
Characteristic velocity		1759.99		m/s
Effective exhaust velocity	2515.02	2771.09	3001.23	m/s
Specific impulse (by mass)	2515.02	2771.09	3001.23	N·s/kg
Specific impulse (by weight)	256.46	282.57	306.04	s
Thrust coefficient	1.4290	1.5745	1.7053	

Table 4. Estimated delivered performance

Parameter	Sea level	Optimum expansion	Vacuum	Unit
Characteristic velocity		1716.41		m/s
Effective exhaust velocity	2364.41	2620.48	2850.63	m/s
Specific impulse (by mass)	2364.41	2620.48	2850.63	N·s/kg
Specific impulse (by weight)	241.10	267.21	290.68	s
Thrust coefficient	1.3775	1.5267	1.6608	

Ambient condition for optimum expansion: H=5.89 km, p=0.473 atm

Table 5. Altitude performance

Altitude, km	Pressure, atm	Effective exhaust velocity, m/s	Specific impulse (by weight), s	Thrust coefficient
0.000	1.00000	2515.016	256.460	1.4290

RPA Calculations:

1) Mass Flow rate of RPA = mass flux at throat x Area of cross-section at throat ;
= 3.2734 kg/s mass flux at throat = 1667.1492 kg/(m²·s)
Area of cross-section at throat = 1963.495 mm^2

2) Thrust (F) = Isp x mass flow rate of RPA x g ; where, Isp = Specific Impulse = 241.1 s (data taken from RPA)
g = 9.81 m/s^2

3) Reaction Efficiency = 97.52 %
Nozzle Efficiency = 97.39 %

Overall Efficiency = Reaction Efficiency x Nozzle Efficiency
= 94.98 %

5. COMPUTATIONAL FLUID DYNAMICS (CFD) ANALYSIS

5.1 Solver Formulation & Governing Equations

A transient, 2D-axisymmetric, pressure-based solver formulation was executed in ANSYS Fluent to resolve the compressible internal gas dynamics and external plume structure. The flow is governed by the time-dependent Reynolds-Averaged Navier-Stokes (RANS) system that comprises the conservation of mass, momentum, and energy:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau}$$

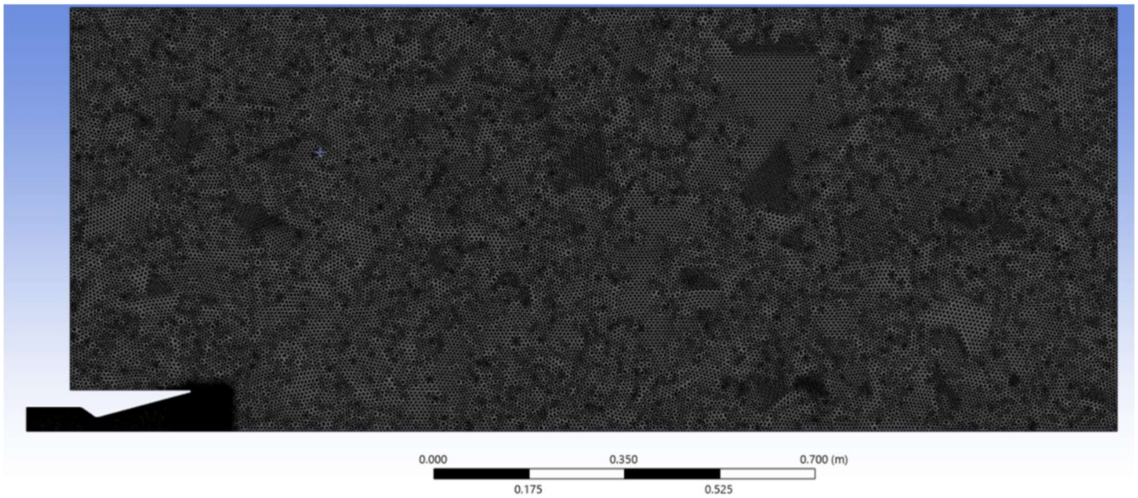
$$\frac{\partial (\rho E)}{\partial t} + \nabla \cdot \big(\mathbf{u}(\rho E + p)\big) = \nabla \cdot (k_{\text{eff}} \nabla T + \mathbf{u} \cdot \boldsymbol{\tau}_{\text{eff}})$$

Turbulence closure was achieved using the k - ω SST (Shear Stress Transport) model to accurately predict adverse pressure gradients, shock-induced boundary layer separation, and shock diamond formations. To resolve high-speed compressible flow phenomena accurately, the model was augmented with the Compressibility Effects Correction (Used to enhanced turbulence dissipation and reduced shear layer mixing in supersonic flow), the Production Limiter (to prevent unphysical over-prediction of turbulent kinetic energy near stagnation regions and sharp geometric corners), and the Viscous Work (viscous heating) was explicitly enabled in the energy equation for turbulent kinetic energy dissipation and viscous dissipation within the high-velocity boundary layers.

5.2 Computational Domain

A 2000 mm × 700 mm domain was used to model the surrounding far-field region.

5.3 Mesh Generation & Mesh Results



Mesh Type	Triangular Mesh Method
Total Number of Nodes	58962
Total Number of Elements	116336
Max Skewness	0.645
Min Orthogonal Quality	0.61363
Max Aspect ratio	2.7

5.4 Material Properties & Operating Conditions

Material Properties

Parameter	**Specification**	**Value**	**Unit**
Working Fluid	LOX & RP-1 Combustion Exhaust	-	-
Density Model	Ideal Gas	-	Kg/m3
Molecular Weight	Constant	23.8185	g/mol
Specific Heat Capacity at Constant Pressure (Cp)	Piecewise Linear Cp (T)	Extracted from RPA results	J/kg-K
Thermal Conductivity (k)	Kinetic Theory (Chapman-Enskog)	-	W/m-K
Viscosity (μ)	Kinetic Theory (Chapman-Enskog)	-	N-s/m2
L-J Characteristic Length (σ)	Kinetic Theory Parameter	3.3	Angstrom
L-J Energy Parameter (ϵ / k_B)	Kinetic Theory (Chapman-Enskog)	264.3	K

Boundary Conditions

Boundary Type	Parameter	Value	Unit
Inlet	Total Pressure	30	bar
	Total Temperature	3567.66	K
	Turbulent Intensity	2	%
	Turbulent Viscosity Ratio	10	-
Outlet	Total Pressure	1.01325	bar
	Prevent Reverse Flow	On	-
	Acoustic Wave Model	Non-Reflecting	-
Wall	Stationary Wall, No-Slip Condition ($\mathbf{u} = 0$)	-	-
	Thermal: Adiabatic / Coupled Heat Transfer	-	-
Axis	2D Axisymmetric Boundary ($y = 0$)	-	-

5.5 Solution Methods & Discretization Schemes

Parameter	Selection / Numerical Scheme	Iteration
Pressure-Velocity Coupling	Coupled Scheme	Entire Simulation
Flux Type	Rhie-Chow: Momentum Based	Entire Simulation
Gradient Discretization	Green-Gauss Node Based	Entire Simulation
Gradient Limiter	Differentiable (Cell to Face Limiting)	Entire Simulation
Pressure Discretization	Second-Order	Entire Simulation
Density, Momentum, & Energy	1. First-Order Upwind 2. Second-Order Upwind 3. Third-Order MUSCL	$\sim$ 1200 time steps (Initial Convergence) $\sim$ 700 time steps (Intermediate Accuracy) $\sim$ 600 time steps (Final Solution Accuracy)
Turbulent Kinetic Energy (k) & Specific Dissipation Rate (ω)	1. First-Order Upwind 2. Second-Order Upwind	$\sim$ 1200 time steps $\sim$ 1300 time steps

Transient Formulation	1. First-Order Implicit 2. Bounded Second-Order Implicit	~ 1200 time steps ~ 1300 time steps
Time Step Size	5e-06 s (5 μs)	Constant (Entire Simulation)
Max Iterations per Time Step	20	Inner loop convergence limit

5.6 Solution Controls & Convergence Parameters

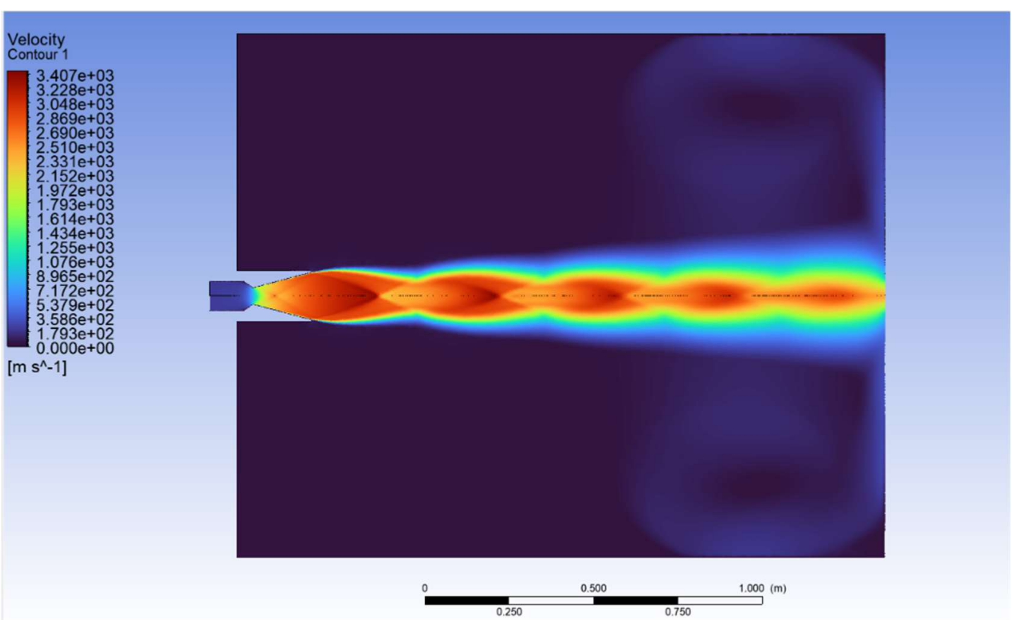
These all settings valid for entire simulation

Parameter	**Selection / Numeric Value**
Flow Courant Number	5
Explicit Relaxation Factor	
Pressure	0.50
Momentum	0.50
Under Relaxation Factor	
Density	1.00
Turbulent Kinetic Energy (k)	0.40
Specific Dissipation Rate (ω)	0.40
Turbulent Viscosity (μ_t)	0.50
Energy	1.00
Residual Convergence Criteria	
Absolute Convergence Criteria (All Equations)	1e-05

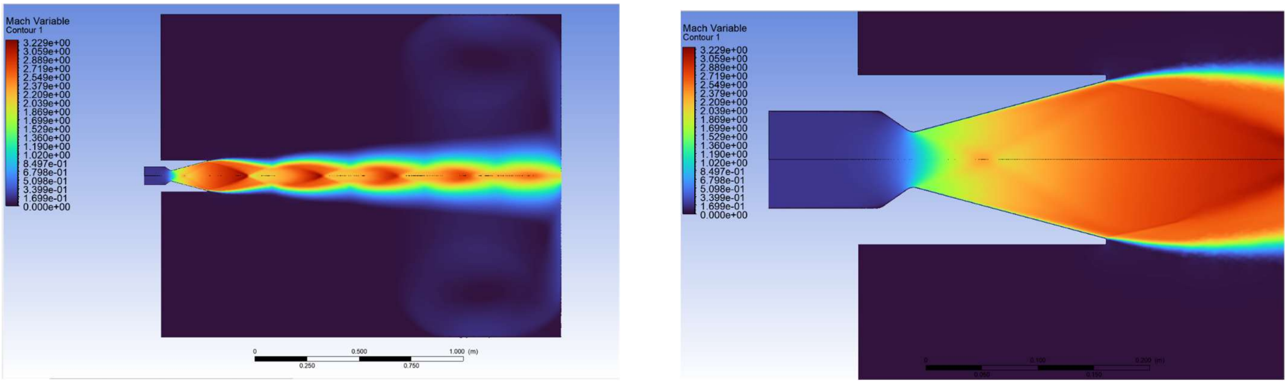
5.7 CFD Results

5.7.1 Flow Field Property Distributions

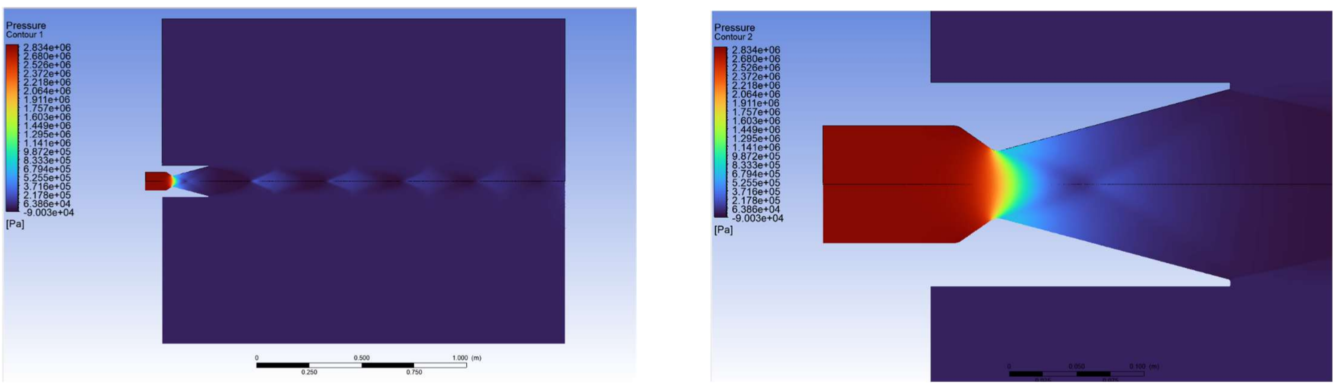
Velocity Contour & Exhaust Gas Dynamics



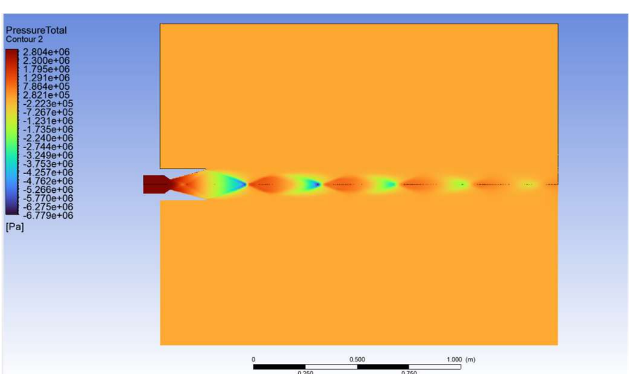
Mach Number Field & Shock Plume Structure



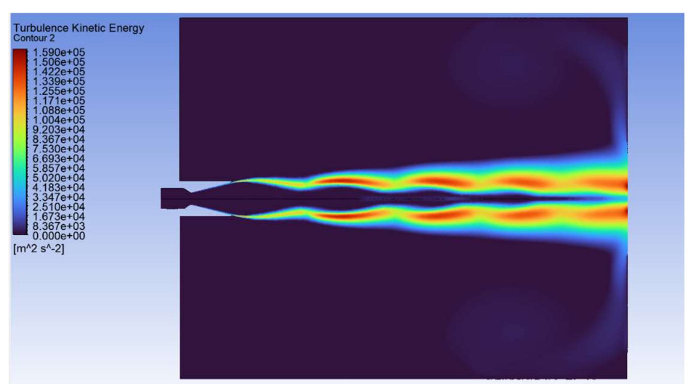
Static Gauge Pressure Distribution



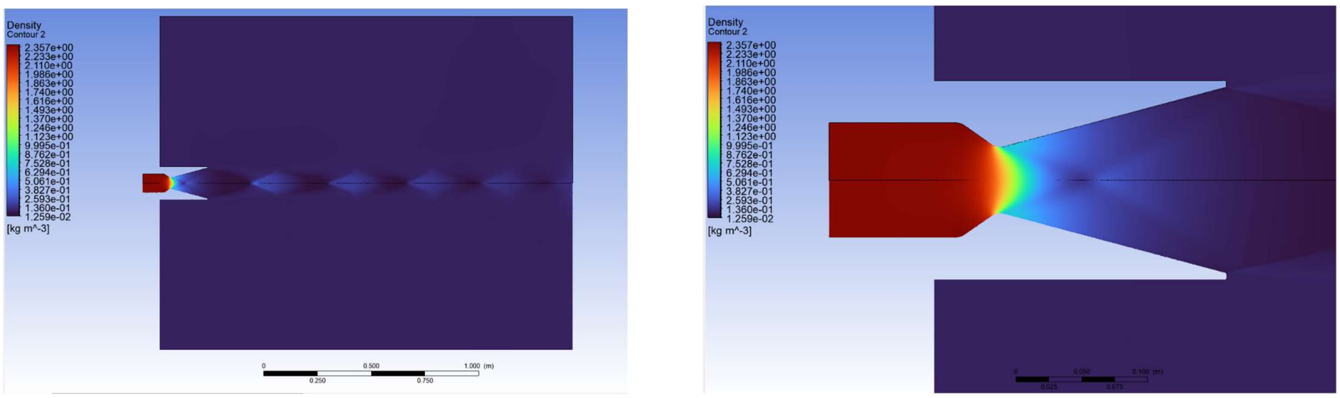
Total Gauge Pressure Distribution

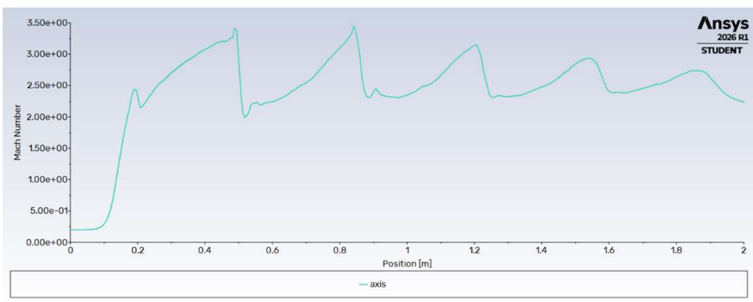


Turbulent Kinetic Energy



Density Distribution





Mach Number Variation along nozzle axial direction

5.7.2 Summary of Flow Field Parameters at Nozzle Throat

Parameter	Value	Unit
Velocity	1165.49	m/s
Static Absolute Pressure	15.175	bar
Mach Number	0.986268	-
Temperature	3467.05	K
Density	1.25216	kg/m3

Calculation:

1) Speed of Sound at nozzle throat (a) = Flow Velocity at throat / Mach Number at throat = 1165.49 / 0.9863
= 1181.717 m/s

5.7.3 Summary of Flow Field Parameters at Nozzle Exit

Parameter	Value	Unit
Velocity	2247.48	m/s
Static Absolute Pressure	0.8014	bar
Mach Number	2.05363	-
Temperature	3092.43	K
Density	0.0741	kg/m3

Calculation:

1) Speed of Sound at nozzle exit (a) = Flow Velocity at exit / Mach Number at exit = 2247.48 / 2.05363 = 1094.394 m/s

5.7.4 Propulsion Performance Parameters & Analytical Calculations

1) Mass flow Rate ($\dot{m}$) = 3.177 kg/s (obtained from CFD)

2) Total Thrust (F) = $\dot{m}$ × Vj + (Pe – Pa) *Ae ; where, $\dot{m}$ = mass flow rate = 3.177 kg/s

Vj = Exhaust Jet Velocity at exit = 2247.48 m/s

Pe = Static Absolute Pressure at exit = 0.801415 bar

Pa = Ambient Pressure = 1.01325 bar

Ae = Area of Cross-section at nozzle exit = 15707.973 mm^2

F = 6807.494 N = 6.8075 kN

3) Specific Impulse (I_{sp}) = F / $\dot{m}$ = 2142.74 N-s/kg
{ By Mass}

Specific Impulse (I_{sp}) = F / ($\dot{m}$ × g) = 218.42 s
{ By Weight}

4) Coefficient of Discharge (Cd) = mass flow rate by CFD / Ideal mass flow rate (from RPA result)

Where, Ideal mass flow rate (obtained from RPA) = 3.2734 kg/s

Cd = 0.97

5) Thrust Coefficient (C_f) = F / (Pc × At) where, Pc = Chamber Pressure = 30 bar

At = Area of Cross-section at throat = 1963.495 mm^2

$C_f = 1.1557$

6) Characteristic Velocity (C^*) = (Pc × At) / $\dot{m}$

C^* = 1854.1 m/s

7) Nozzle Efficiency = $\left(\frac{Velocity\ of\ exhaust\ by\ CFD}{Ideal\ Velocity\ of\ Exhaust}\right)^2$

Nozzle Efficiency = 88.3 %

Where, Ideal Velocity of Exhaust

$$v_{effective} = \sqrt{\frac{2\gamma}{\gamma - 1} RT_c \left[1 - \left(\frac{P_e}{P_c}\right)^{\frac{\gamma-1}{\gamma}}\right]} = 2647.2634\ m/s$$

γ = Specific Heats Ratio = 1.2287 (from RPA results)

R = Characteristic Gas Constant = 340.4 J/kg-K (from RPA results)

Tc = Chamber Temperature = 3567. 6589 K

Pc = Chamber Pressure = 30 bar

Pe = Exit Pressure = 0.48 bar

$$v_{e,ideal} = v_{effective} + \frac{(P_e - P_a)*Ae}{\dot{m}} = 2391.37 \text{ m/s}$$

6. VALIDATION OF PERFORMANCE ANALYSIS: CFD vs RPA

Parameter	**CFD Result**	**RPA Result**	**Variation (%)**	**Unit**
Exhaust Velocity	2247.48	2391.37	6.02	m/s
Mach Number at throat	0.986	1.00	1.40	-
Mach Number at exit	2.054	3.1304	34.39	-
Throat Static Absolute Pressure	15.175	17.004	10.76	bar
Exit Static Absolute Pressure	0.801415	0.48	66.96	bar
Temperature at Throat	3467.05	3401.8086	1.92	K
Mass flow rate	3.177	3.2734	2.94	kg/s
Total Thrust	6807.494	7742.216	12.07	N
Specific Impulse	218.42	241.1	9.41	S
Thrust Coefficient	1.1557	1.3775	16.1	-
Nozzle Efficiency	88.3	94.98	7.03	%

7. COUPLED THERMAL-STRUCTURAL FINITE ELEMENT ANALYSIS (FEA)

7.1 Objective & Governing Equations

Evaluate the thermal gradients, structural deformation, and combined thermo-mechanical stresses act on the nozzle wall under high-temperature combustion and internal static pressure loads.

Thermal Conduction (Fourier's Law): $q = -k\nabla T$

Thermal Expansion Strain ($\varepsilon_{\text{thermal}}$): $\varepsilon_{\text{thermal}} = \alpha(T - T_{\text{ref}})$

Total Stress (Hooke's Law with Thermal Strain): $\sigma = \mathrm{C} : (\varepsilon_{\text{total}} - \varepsilon_{\text{thermal}})$

Equivalent Stress (von Mises stress): $\sigma_v = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}$

Where, σ_1, σ_2, σ_3 are principal stresses in x, y, z direction respectively.

k = Thermal Conductivity

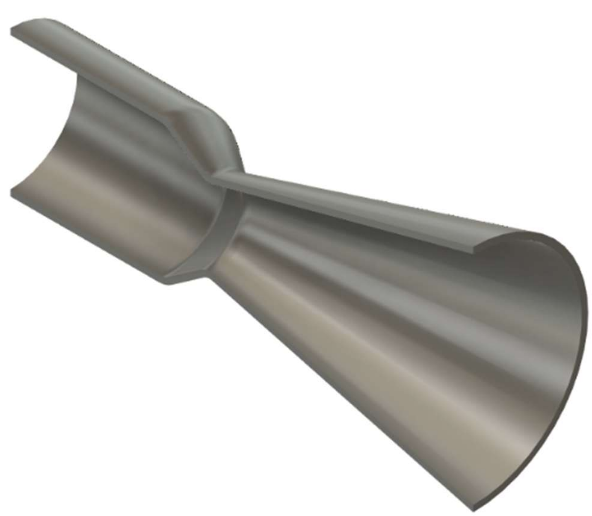
α = Coeff. of Thermal Expansion

C = Elasticity Tensor

T_{ref} = Zero-stress reference Temperature

7.2 3D CAD Design

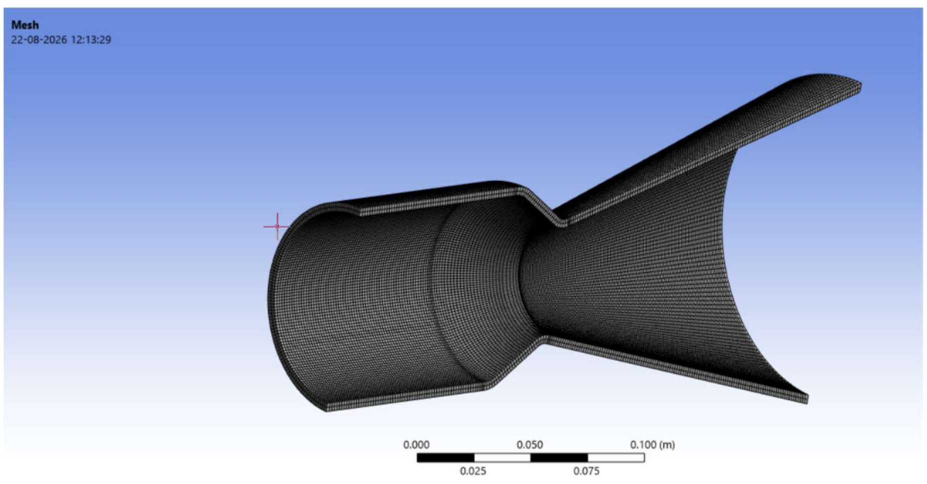
Geometrical Dimensions are same as CFD (except Surrounding domain)
Thickness = 4mm



7.3 Engineering Materials & Properties

Property	Dependent	Value	Unit
Material	-	Inconel 718	-
Density	Temperature Dependent	7300 to 8220	Kg/m3
Young's Modulus	Temperature Dependent	34 to 165	GPa
Poisson's Ratio	Temperature Dependent	0.28 to 0.4	-
Thermal Conductivity	Temperature Dependent	11 to 25	W/(m-K)
Coeff. of Thermal Expansion	Temperature Dependent	1.4e-05 to 1.85e-05	per °C
Yield Strength	Temperature Dependent	34 to 648	MPa

7.4 Mesh Generation & Mesh Results



Mesh Type	Sweep Mesh Method
Total Number of Nodes	67410
Total Number of Elements	44304
Maximum Skewness	0.25709
Maximum Aspect Ratio	3.3377

7.5 Thermal Analysis & Results

7.5.1 Applied Conditions:

1) Convection Heat Transfer on inner surface of Nozzle with hg calculated from Bartz Convective Heat Transfer Coefficient Equation

$h_g = \left[\frac{0.026}{D_t^{0.2}}\left(\frac{\mu_0^{0.2}\,C_p}{\mathrm{Pr}^{0.6}}\right)\left(\frac{P_c}{c^*}\right)^{0.8}\left(\frac{A_t}{A}\right)^{0.9}\right]\sigma*$; where, Dt = Throat Diameter = 50 mm

Pr = Prandtl Number of Exhaust gas = 0.7

Pc = Chamber Pressure = 30 bar

c^* = Characteristic Velocity of exhaust gas = 1854.1 m/s (get from CFD results and Analytical Calculation)

At = throat Area

A = Area of cross-section at any region

$\sigma*$ = Boundary layer correction factor for property variations = 1.0

μ_0 = Dynamic Viscosity of gas at chamber (get from Chapman-Enskog kinetic theory equation)

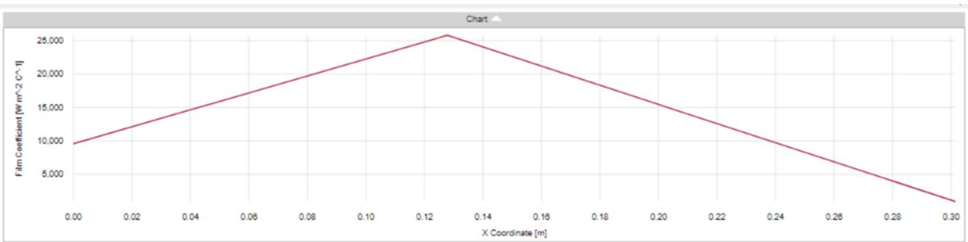
$\mu_0 = 2.6693 \times 10^{-6}\frac{\sqrt{M \cdot T}}{\sigma^2 \cdot \Omega_\mu}$; M = Molecular Mass of gas (g/mol)

T = Absolute Temperature (K)

σ = Lennard-Jones characteristic length (in Angstrom) = 3.3

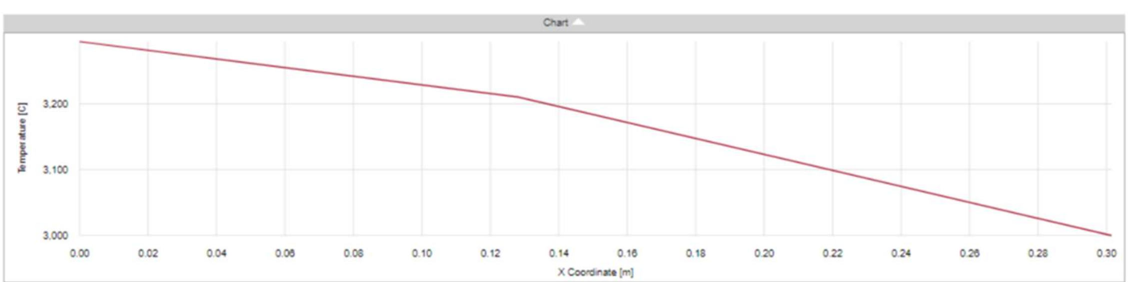
$$\text{(Viscosity Collision Integral) } \Omega_\mu = \frac{1.16145}{(T^*)^{0.14874}} + \frac{0.52487}{\exp(0.77320 \cdot T^*)} + \frac{2.16178}{\exp(2.43787 \cdot T^*)}$$

where, $T^* = \frac{k_B T}{\epsilon}$; ϵ/k_B = Lennard-Jones Energy Parameter = 264.3 K



Row	X Coordinate [m]	Film Coefficient [W m^-2 C^-1]
1	0	9556.75
2	0.127837	25782.309
3	0.301482	925.495
*		

Here hg is plotted against nozzle axial length and key points data is marked at inlet, throat, exit

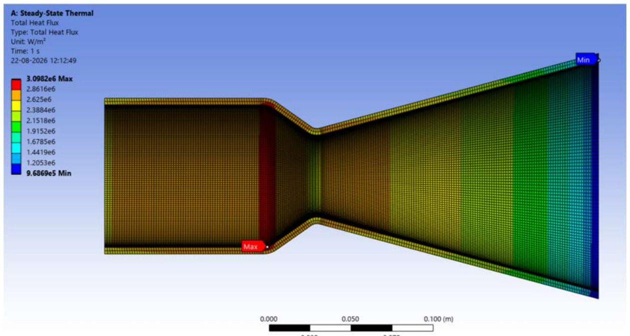


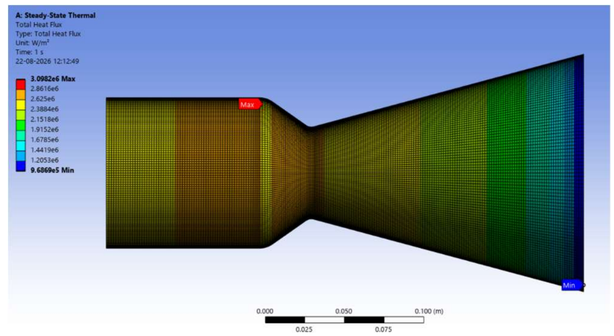
Row	X Coordinate [m]	Temperature [C]
1	0	3294.5089
2	0.127837	3210.54
3	0.301482	2999.86
*		

2) Radiation on Outer Surface of Nozzle with emissivity = 0.8 and to ambient temperature (22 °C)

7.5.2 Results:

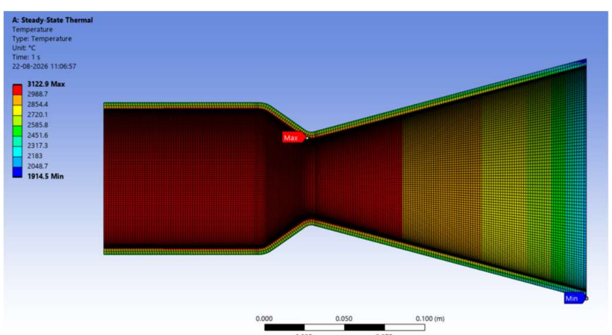
Total Heat Flux

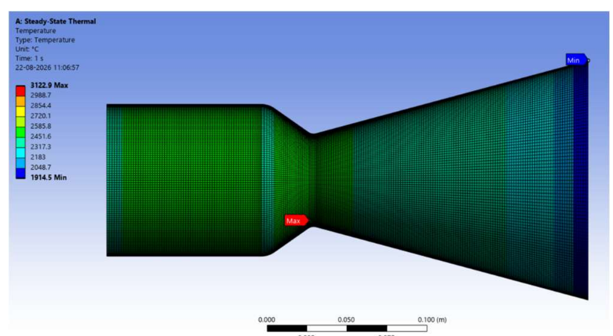




Maximum Total Heat Flux = 3.0982 MW/m2 (at starting of converging section of nozzle)
Minimum Total Heat Flux = 0.969 MW/m2 (near nozzle exit)

Temperature

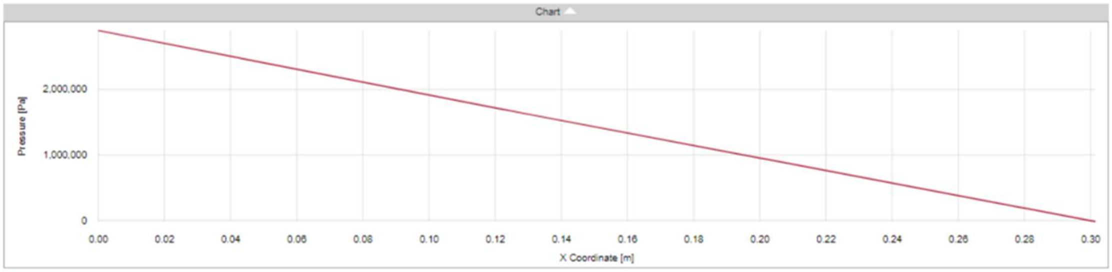




Maximum Temperature = 3396.05 K (near throat)
Minimum Temperature = 1914.5 K (near nozzle exit)

Inconel 718 melting temperature is nearly 1600 K but here maximum temperature reached 3396.05 K without any external support of cooling. That's why regenerative cooling like film cooling, and impingement jet cooling techniques used to cool the surface of nozzle.

7.6 Structural Analysis & Results



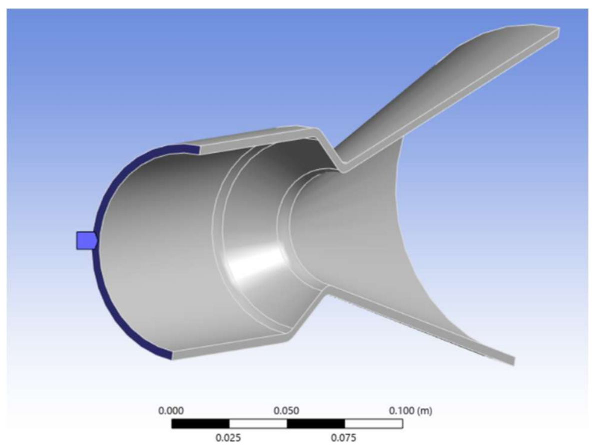
Row	X Coordinate [m]	Pressure [Pa]
1	0	2898675
2	0.127837	1641610
3	0.301482	-17696.2
*		

7.6.1 Applied Conditions:

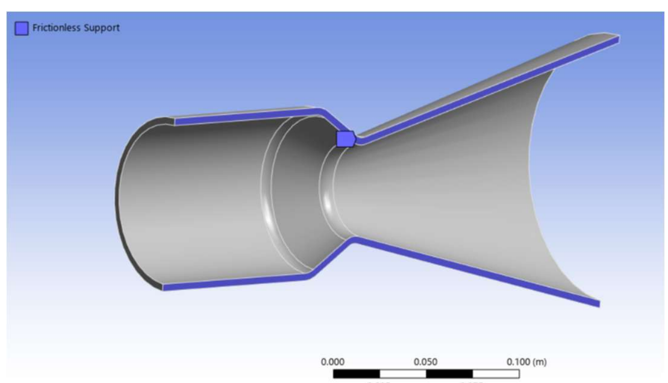
1) Normal Pressure is applied on inner surface of nozzle

Here static gauge pressure is plotted against nozzle axial length and key points data is marked at inlet, throat, exit

2) Fixed Support at inlet flange



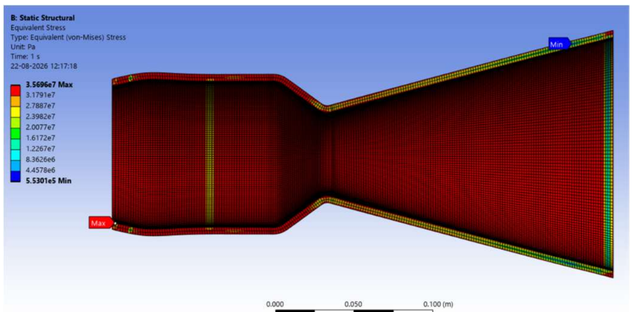
3) Frictionless Support at axial cross-section lengths

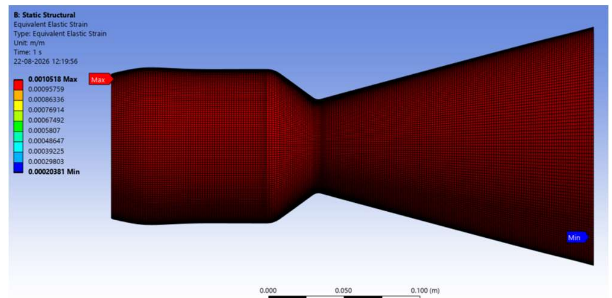


4) Imported Thermal Load from Thermal Analysis results to evaluate total stress and strain

7.6.2 Results:

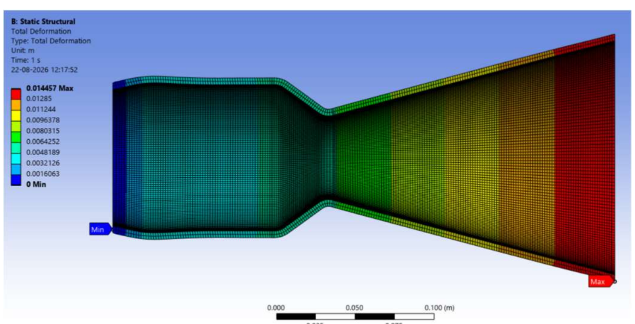
Equivalent (Von-Mises) Stress

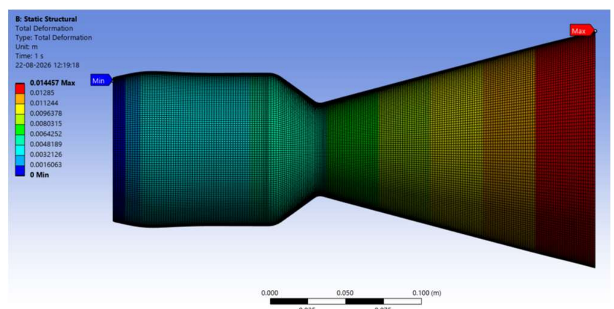




Maximum Von-Mises Stress = 35.696 MPa (near inlet)

Total Deformation

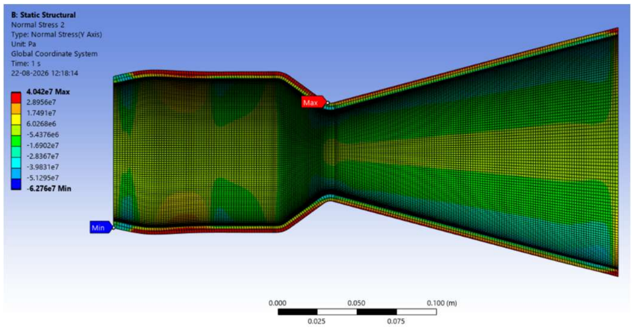


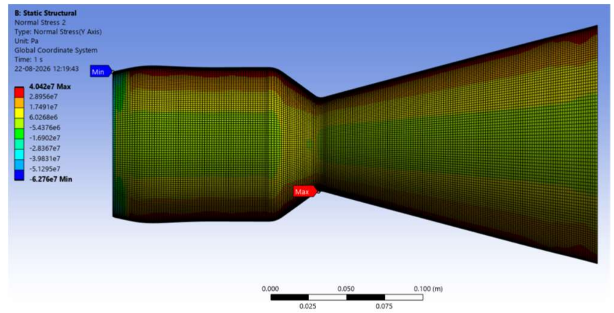


Maximum Total Deformation = 0.014457 m (at exit)

Minimum Factor of Safety (FoS)min = Yield Strength / Max. Von-Mises Stress = 1.0358

Hoop Stress (along nozzle thickness)





Maximum Hoop Stress = 40.42 MPa (at throat)

8. SUMMARY OF PERFORMANCE OUTCOMES & MULTIPHYSICS RESULTS

S.No.	Parameter	Value	Unit
1.	Thrust	6807.494	N
2.	Specific Impulse (by Weight)	218.42	s
3.	Exhaust Velocity (Vj)	2247.48	m/s
4.	Mach Number at nozzle exit	2.05363	-
5.	Mach Number at nozzle throat	0.9863	-
6.	Ac/At (Chamber Area / throat Area)	3.0	-
7.	Ae/At (Exit Area /throat Area)	8.0	-
8.	Coefficient of Discharge (Cd)	0.97	-
9.	Thrust Coefficient (Cf)	1.1557	-
10.	Characteristic Velocity (c^*)	1854.1	m/s
11.	Nozzle Efficiency	88.3	%
12.	Maximum Heat Flux	3.0982	MW/m^2
13.	Maximum Wall Temperature	3396.05	K
14.	Maximum Von-Mises Stress	35.696	MPa
15.	Maximum Total Deformation	14.457	mm

16.	Minimum Factor of Safety (FoS)min	1.0358	-

9. CONCLUSION

The main goal of this project was to analyze the gas dynamics, thermal behaviour, and structural response of an Inconel 718 convergent-divergent rocket nozzle using a coupled CFD-FEA approach. By running 2D compressible CFD simulations, thermal loading, and transferring the resulting thermal loads into a steady static-structural FEA model, we were able to map out how the nozzle performs under typical operating pressures and temperatures.

10. LEARNING OUTCOMES

This project provided deep practical experience in high-speed gas dynamics, thermal boundary layer theory, and structural mechanics. Through the implementation of analytical correlations including the Bartz equation, and Chapman-Enskog kinetic theory for viscosity, I gained a solid understanding of how real-world rocket nozzle parameters and scaling laws operate.

By building an end-to-end 1-Way coupled multiphysics workflow, I developed strong technical skills in setup and mesh refinement for 2D compressible CFD to resolve supersonic flow phenomena, exit Mach numbers, and non-dimensional metrics (Cd, Cf, nozzle efficiency). Transferring those CFD pressure and heat fluxes data into a solid FEA model shows my capability to analyze directional thermal expansion, evaluate throat stress concentrations, and determine structural Factors of Safety against high-temperature Inconel 718 yield criteria.

11. FUTURE OBJECTIVES

Moving forward, I plan to run a full parametric study to see how this nozzle profile behaves across different flight altitudes, specifically looking at the flow physics in overexpanded, optimum, and underexpanded conditions. The goal is to tweak the contour so it stays as close to optimum expansion as possible over a wider altitude range to boost overall thrust efficiency. Beyond geometry tweaks, I want to add regenerative cooling channels into the structural mesh to control the extreme throat temperatures, and eventually explore igniter dynamics (like hot-gas torch igniters) to see how startup transients affect the thermal shock on the wall.

12. REFERENCES

1) Numerical Simulation of Supersonic Flow Through a Convergent-Divergent Nozzle, Ekanayake, E. M. Sudharshani, B.Aero Eng.(Hons), RMIT University.

2) A Simple Equation for Rapid Estimation of Rocket Nozzle Convective Heat Transfer Coefficients, D. R. BARTZ, Jet Propulsion Laboratory, California Institute of Technology.

3) Thermal Analysis and Design Optimization of Inconel 718 Convergent -Divergent Nozzle for High Performance Propulsion System, Raja Munusamy, Alliance University.